

Signal Processing Project

Report · Q1A & Q1B

Q1A: Frequency Forensics: The Ghost Signal

1. 2D DFT and its Inverse

For an image $I(x,y)$ of size $M \times N$ pixels, the 2D DFT breaks the image down into a sum of 2D sine/cosine waves of different spatial frequencies (u,v) . The forward transform is:

$$F(u,v) = \sum_{x=0..M-1} \sum_{y=0..N-1} I(x,y) e^{-j2\pi(ux/M + vy/N)} \quad (\text{forward DFT})$$

The inverse just reverses this — it adds all those waves back together, weighted by $F(u,v)$, to rebuild the original image:

$$I(x,y) = (1/MN) \sum_{u=0..M-1} \sum_{v=0..N-1} F(u,v) e^{+j2\pi(ux/M + vy/N)} \quad (\text{inverse DFT})$$

Is the spectrum naturally centred? No. When NumPy computes the DFT, it stores the zero-frequency term $F(0,0)$, the DC component, i.e., the image's average brightness at the top-left corner, index $[0,0]$, with frequency increasing outward from there in a wraparound order. That's just a storage convention, not anything physical about the image. To view the spectrum the way we normally think of one — zero frequency in the middle, increasing symmetrically out to the edges in every direction, we apply `fftshift`, which simply rearranges the array so the DC term lands at the centre. We use this shifted, centred version for every spectrum plot in this report.

2. Corrupted Image and Spectrum

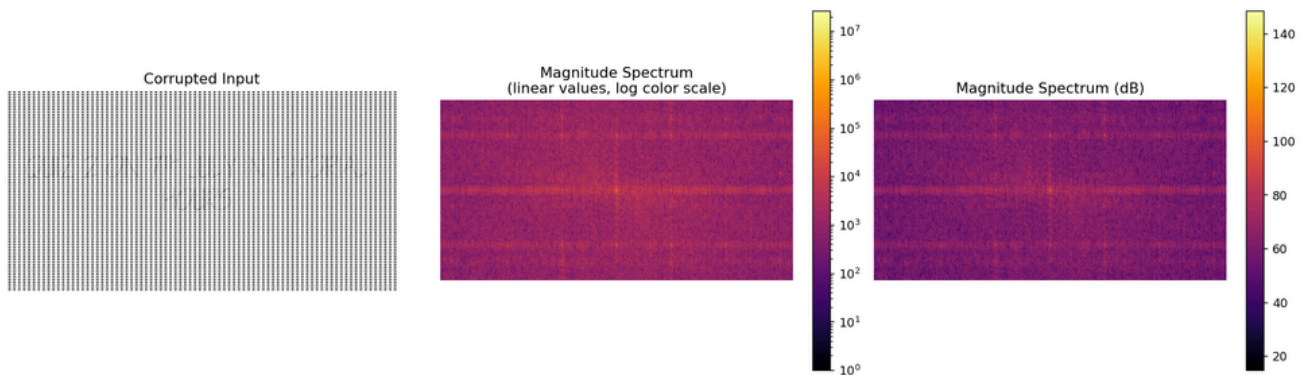


Figure 1: Corrupted input, magnitude spectrum (linear values, log color scale for visibility), and magnitude spectrum in dB.

In both the linear-scale and the dB-scale spectrum, the single brightest point sits exactly at the centre, which is the DC component (the image's average brightness), and it dominates because most of the image is a fairly uniform background. Scattered around it, in a regular grid pattern, are dozens of much smaller but still sharp bright dots at fixed off-centre positions. **These are the dominant components responsible for the corruption.** A signal that repeats periodically in space always produces a DFT made of a few sharp spikes (at the repetition frequency and its harmonics), never a smooth spread, so a periodic pattern of dots in the image becomes a periodic pattern of spikes in the spectrum. That's the key difference from genuine image content: real content (edges, shading, text strokes) does not repeat on a fixed short cycle, so it spreads its energy smoothly across many frequencies instead of concentrating it in a few isolated spikes.

3. Locating the Noise: Peak Detection

Rather than picking an arbitrary number of peaks, the noise floor of the genuine image-content spectrum was measured (99.9th percentile = 85.8 dB), and a 2 dB margin was added as a threshold (87.8 dB). Every pixel above this threshold, away from the DC region, is treated as a noise impulse. 120 such impulses were found, forming the regular lattice below.

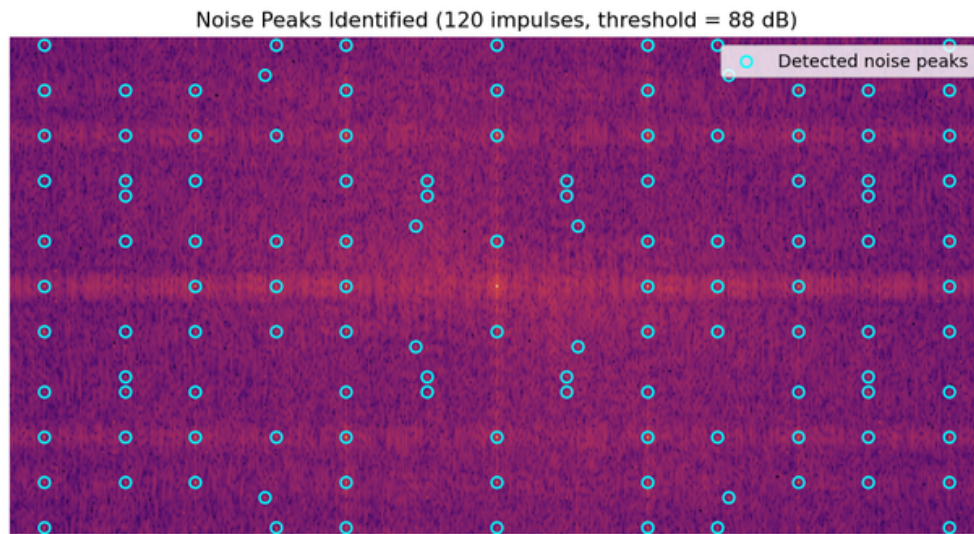


Figure 2: Detected noise impulses (cyan circles) overlaid on the dB spectrum.

Why these particular frequencies? Their position in the spectrum is set directly by how often the dot pattern physically repeats across the image. This isn't an arbitrary pick; it can be checked directly against the corrupted image itself.

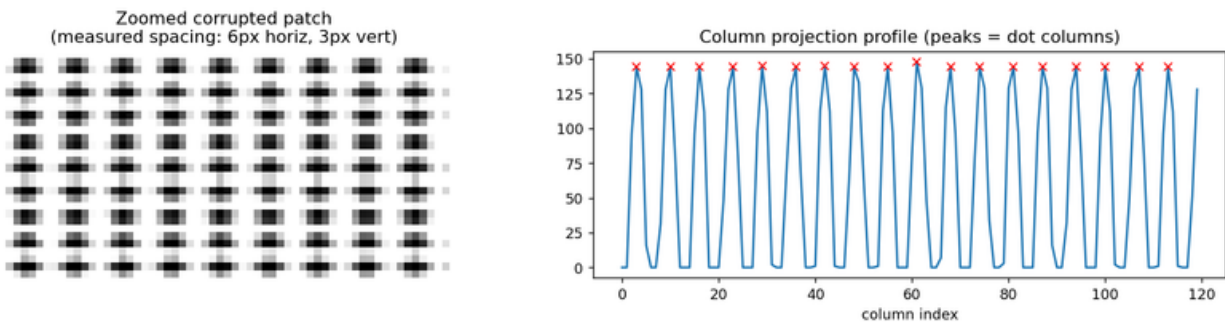


Figure 2a: Measuring the dot grid's spacing directly from the corrupted image via column/row projection profiles.

Counting dot-to-dot spacing directly in the image gives a horizontal period of 6.0 px and a vertical period of 3.0 px. The DFT relates a spatial period T (pixels) to a spectral peak at index N/T , so for this image ($W=517$, $H=264$), that predicts peaks near column-index $517/6.0 \approx 86$ and row-index $264/3.0 = 88$. The strongest peaks actually detected sit at index 80 in both directions, close to the prediction, with the small gap expected because the dot grid isn't a perfect sinusoid (its energy spreads slightly around the fundamental rather than landing on one exact bin). This match is the justification for notching exactly these locations: they are precisely where the visible repetition rate of the corruption says they should be, not a guess.

Property	Noise peaks	Image content
Location	Discrete, on a regular lattice	Spread broadly, concentrated near DC
Shape	Sharp, isolated impulses	Smooth, continuous fall-off
Magnitude	Above 87.8 dB	Below the 99.9th-percentile content ceiling

4. Notch Filtering and Reconstruction

A circular notch (radius 5 px, Gaussian-smoothed edges to avoid ringing) is placed at every detected peak, and the masked spectrum is inverted back to the spatial domain with the IDFT.

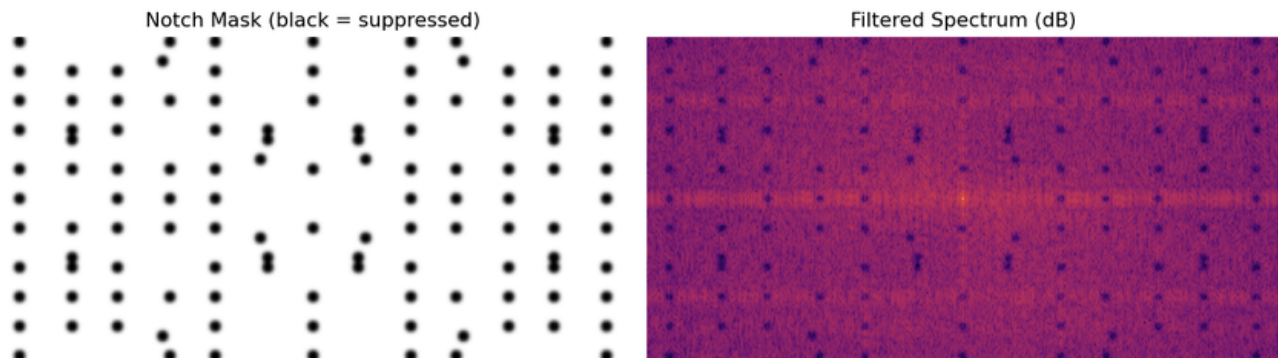


Figure 3: Notch filter mask and the resulting filtered spectrum.



Figure 4: Corrupted input, raw IDFT output, and the same recovery after a percentile-based contrast stretch (display only the underlying recovered signal is unchanged; the raw values just span a narrow ~115-215 band out of 0-255).

The hidden message has been successfully recovered: HIDDEN MESSAGE SUCCESSFULLY RECOVERED."

Quantitatively, the detected noise-peak bins sat 37.9 dB above the background spectrum before filtering, and -4.7 dB (i.e. below the natural floor) after confirming the periodic interference was removed, not just attenuated.

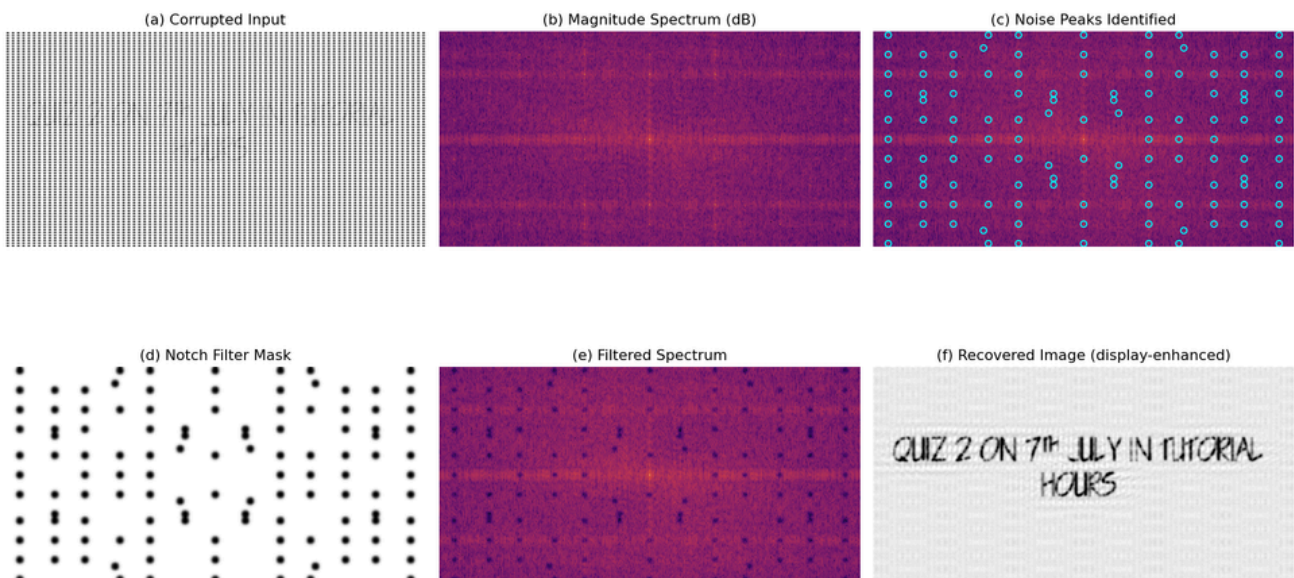


Figure 5: Complete recovery pipeline — (a) corrupted input, (b) spectrum, (c) detected peaks, (d) notch mask, (e) filtered spectrum, (f) recovered image.

5. Effect of Filter Parameters



Figure 6: Reconstruction at notch radius 2, 5, 12, and 25 px.

Notch radius	Effect
2 px	Too small grid pattern still clearly visible
5 px (used)	Clean message fully legible, no residual grid
12 px	Mild blur some genuine high-frequency detail starts to go
25 px	Destroys the image encroaches on low-frequency content near DC

Removing more frequencies does not always help. Past a point, the notch starts deleting genuine image content rather than noise. The best filter is the smallest one that fully suppresses every detected peak not the largest one available.

6. Practical Applications

- **Communications:** removing narrowband jamming/interference from received image data.
- **Remote sensing:** eliminating scan-line or sensor-banding noise from satellite imagery.
- **Surveillance:** recovering readable footage corrupted by RF interference.
- **Medical imaging (MRI/CT):** removing periodic artifacts from motion or equipment vibration.

Q1B: Digital Detective: Missing Boundaries

1. Gradients and the Sobel Filter

An edge is a location of large intensity gradient: $|\nabla I| = \sqrt{(\partial I / \partial x)^2 + (\partial I / \partial y)^2}$. The Sobel operator approximates this with two 3x3 convolution kernels (K_x for vertical edges, K_y for horizontal edges); the ± 2 centre weights give a built-in smoothing that reduces pure-derivative noise sensitivity.

2. Sobel Gradient Components

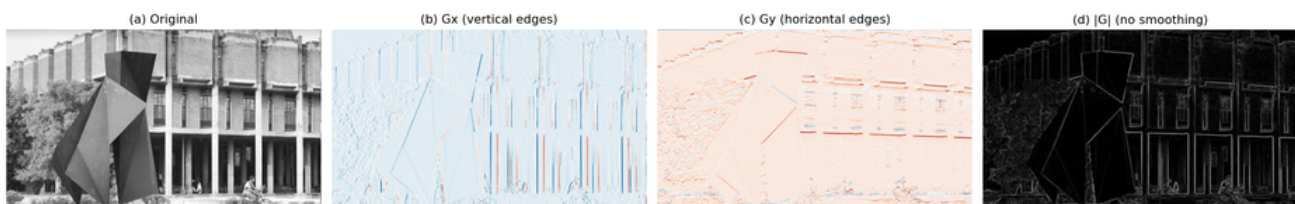


Figure 7: (a) Original campus photo a brick building facade with regular window/column lines, a faceted steel sculpture, trees, and a cyclist. (b) G_x highlights vertical lines (columns, sculpture edges). (c) G_y highlights horizontal lines (window ledges, roofline). (d) Combined magnitude $|G|$.

3. Role of Pre-smoothing

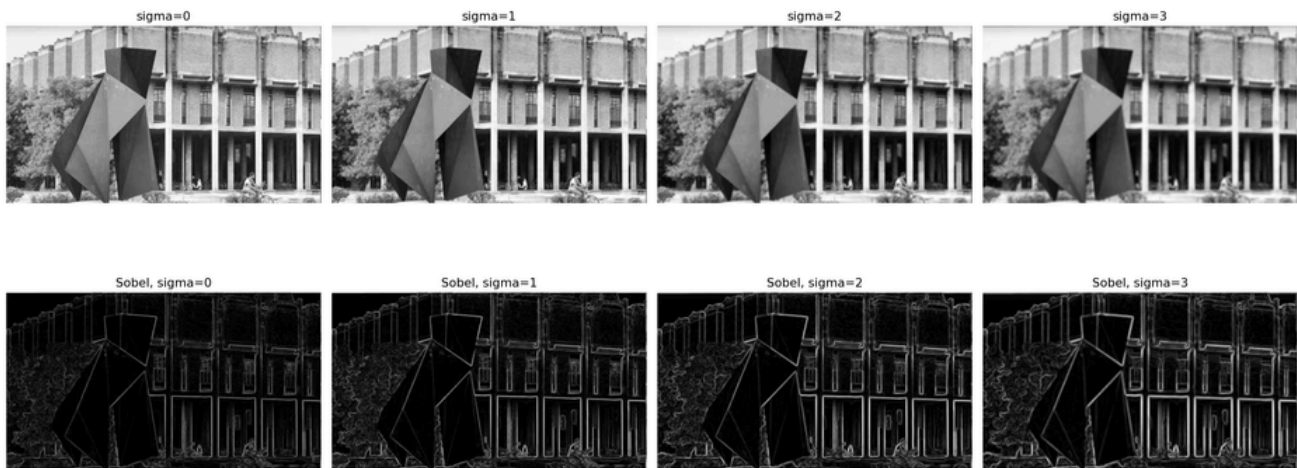


Figure 8: Effect of Gaussian pre-smoothing ($\sigma = 0, 1, 2, 3$) before applying Sobel.

Without smoothing, fine texture and sensor noise produce many spurious edges. Gaussian smoothing acts as a low-pass filter before differentiation, equivalent to convolving with a derivative-of-Gaussian, the basis of the Canny edge detector.

4. Noise Robustness

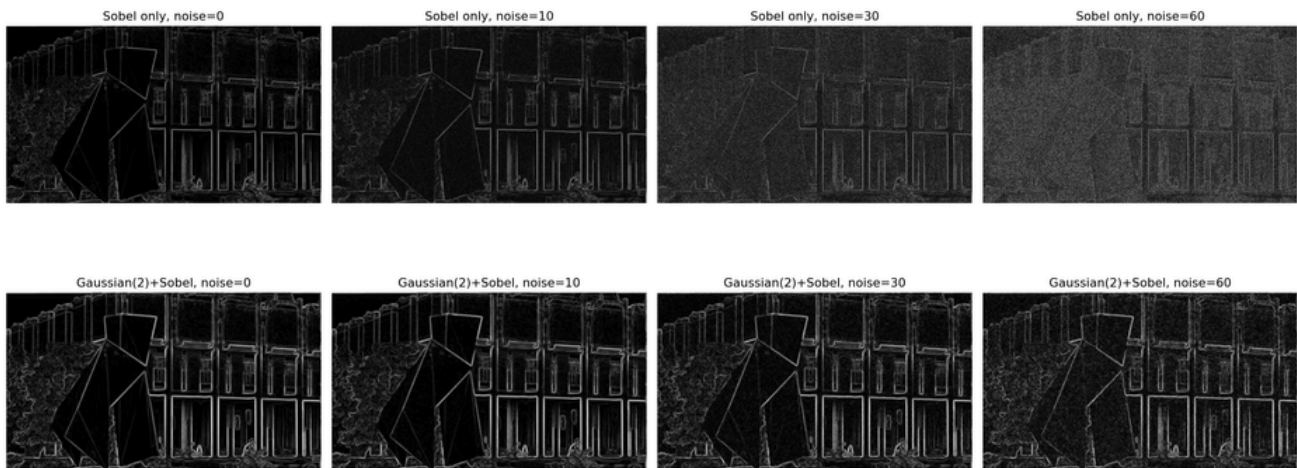


Figure 9: Sobel-only vs. Gaussian($\sigma=2$)+Sobel at noise levels $\sigma = 0, 10, 30, 60$.

Sobel reacts to any rapid intensity change, so noise produces edges just as strong as real boundaries. At noise=30-60, the Sobel-only output is dominated by speckle, while the Gaussian-smoothed version still shows the building's window grid clearly.

Gaussian σ	Noise suppression	Weak-edge preservation	Localization
0	None	Best	Pixel-precise
1	Mild	Good	Good
2 (used)	Strong	Moderate	Moderate
3	Very strong	Poor	Blurred / shifted

Weak edges without excess noise: moderate smoothing ($\sigma \approx 2$) combined with a modest threshold ($\sim 30\%$ of max) keeps faint real edges (foliage, distant railings) while rejecting noise; too high a threshold erases the faint ones instead. $\sigma=2$ is the sweet spot for this image: noise is well suppressed while the window grid and sculpture facets stay distinct.

5. Final Result

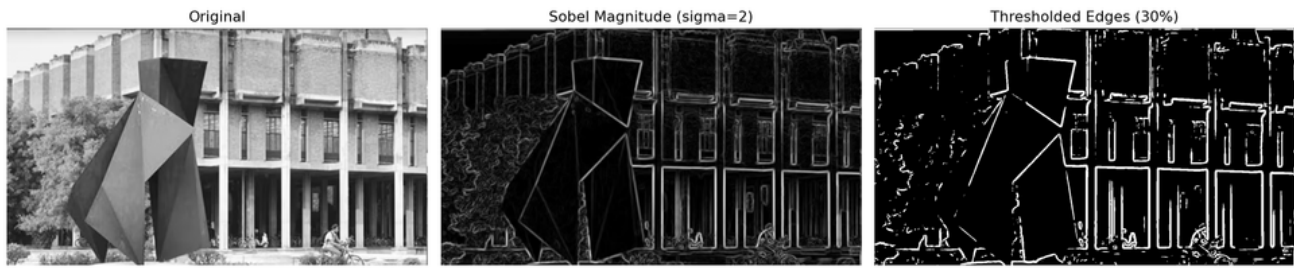


Figure 10: Original, Sobel magnitude ($\sigma=2$), and thresholded binary edges (30% of max).

Conclusion

Both systems work as intended on the provided inputs. Q1A recovered the hidden text **the hidden message**

by identifying and notch-filtering the periodic noise lattice in the frequency domain. Q1B revealed the structural boundaries of the campus building/sculpture scene using Sobel gradients, with Gaussian pre-smoothing ($\sigma=2$) giving the best noise-vs-detail trade-off.